

Program : Diploma in Instrumentation Engineering	
Course Code : 4089	Course Title: Maintenance & Calibration Workshop II
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on experience with wiring, tagging, tube bending & fitting
- To familiarize calibration works needed in instrumentation job sites

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of 5S to be followed in workshop		Maintenance & Calibration Workshop I	2
Knowledge of Electrical Parameter measurement		Maintenance & Calibration Workshop I	2
Knowledge of active and passive components used in networks		Maintenance & Calibration Workshop I	2

Course Outcomes:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Implement Instrument wiring & tagging and relay logic circuits for basic control applications	9	Applying
CO2	Implement Instrument fitting, tubing and threading	15	Applying
CO3	Perform cable glanding, shrouding and troubleshooting	9	Applying
CO4	Perform servicing of pressure, temperature and level measuring instruments	9	Applying
	Lab Test	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		2			2	
CO2	3			2			
CO3	3	2		2			
CO4	3					2	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Implement Instrument wiring & tagging and relay logic circuits for basic control applications		
M1.1	Instrument wiring & tagging: i) Study and Practice Loop wiring & panel wiring and Junction Box Wiring. ii) Practice Wiring of instrument panel board with various accessories. iii) Test and verify assembled instrument loop wiring for various parameters and faults. iv) Understand Instrument Tagging, Marshalling, Cable Termination & Feruling.	3	Applying
M1.2	Relay Logic Practice: i) Study and identify the connections on Relays ii) Implement a circuit to control 230V/60w bulb using 12V /24V Relay iii) Implement a circuit for the start and stop of a 12V DC motor using a relay. iv) Implement a circuit to run a DC motor in Forward Direction & Reverse Direction using DPDT Relay	6	Applying

CO2	Implement Instrument fitting, tubing and threading		
M2.1	Study of Instrument Fittings: i) Study Elbow, Tee, Union, conversion union, reducing union, Union Elbow, Union tee, connectors, nipples used for instrument fitting. ii) Study the different types of Threads – NPT, BSP, BSPT, BSPP etc iii) Study 2 way, 3 way and 5 way Manifolds iv) Study Instrument hook-ups for Liquid, Gas & Steam pipe Lines	6	Applying
M2.2	Tubing and Threading: i) Identify the tools for tubing & threading works - Tube Cutter, Tube Bender Swage tool, Die, Tap etc. ii) Perform Bending, flaring and glanding of metal tubes iii) Practice riveting work on metal sheet	9	Applying
	Lab Test – I	1	
CO3	Perform cable glanding, shrouding and troubleshooting		
M3.1	Cable glanding, Shrouding: i) Identify Signal Cable and Power Cables ii) Identify the fittings for Cable Glanding & Perform Cable glanding & Shrouding	3	Applying
M3.2	Loop Checking: Testing of instrumentation loop, Instrumentation panel and Perform loop checking.	3	Applying
M3.3	Calibration: Familiarization of hand held calibrators - HART and Foundation FIELDBUS	3	Applying
CO4	Perform servicing of pressure, temperature and level measuring instruments		
M4.1	Service and Repair of Pressure gauges: i) Dismantle and assemble Bourdon tube, bellow and diaphragm type pressure gauges ii) Dismantle and assemble Dead weight tester	3	Applying
M4.2	Installation and Testing of Temperature sensors: Study the Installation and testing of different types of temperature sensors	3	Applying

M4.3	Pressure and Level switches: i) Set pressure switches for differential pressure ii) Dismantle, assemble and set Level switches	3	Applying
	Lab Test – II	2	

Text / Reference:

T/R	Book Title/Author
R1	S K Singh , Industrial Instrumentation and Control second edition
R2	Fundamentals of Instrumentation (In partnership with NJATC), second edition, Thomson Delmar Learning
R3	Dr. Jasbir Singh Saini, Textbook of Measurements and Instrumentation (AICTE), New Age International (P) Ltd., Publishers

Online Resources:

Sl.No	Website Link
1	https://instrumentationtools.com/loop-checking/
2	https://instrumentationtools.com/troubleshooting-with-loop-calibrator/
3	https://instrumentationtools.com/loop-checking-4-20-mahart-transmitters/
4	https://instrumentationtools.com/instrument-hook-up-diagram/
5	https://www.youtube.com/watch?v=F-hkFrKhFrw